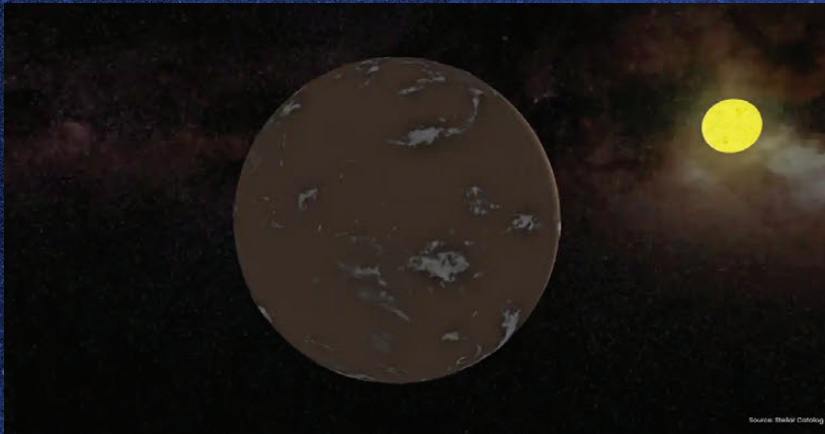




Meet TOI-6651b, a new rare planet 5 times the size of Earth discovered by Ahmedabad scientists

New planet alert! An international team of scientists along with the Physical Research Laboratory (PRL) in Ahmedabad has discovered a new exoplanet. It is being called TOI-6651b and was discovered using a PARAS-2 spectroscope. This high-precision instrument was mounted on PRL's 2.5m telescope at Mount Abu in

Rajasthan. The newly discovered exoplanet is five times the size and 60 times the mass of Earth.

It is a sub-Saturn class planet and is located 690 light-years away from the Sun. For those unaware, sub-Saturn planets fall in the size gap between Neptune and Saturn. TOI-6651b is now added to the list and is only the fourth exoplanet

found within the rare "Neptunian desert". The Neptunian desert is a region close to stars where planets of Neptune's size are rarely seen.

The TOI-6651b too orbits a Sun-like star in just five days. It is exposed to intense radiation that could strip away its gaseous atmosphere, potentially leaving behind a rocky core. Nearly 87% of the exoplanet's core consists of metals like iron, with an outer layer of hydrogen and helium. Its temperatures range to around 1,200°C, so TOI-6651b is far from habitable.

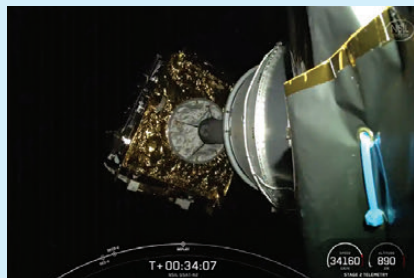
Sanjay Baliwal, lead author and PRL researcher said, "TOI-6651b was located at the edge of the Neptunian desert, which is crucial for understanding factors that shape the desert boundaries." Speaking about the PARAS-2 spectroscope, he said, "Its high resolution enabled the detection of this sub-Saturn exoplanet, which would not have been possible with its predecessor."

This discovery is now published in the Astronomy and Astrophysics journal which helps get an insight on the evolution of planets in extreme environments and offers clues to the role of tidal forces in shaping planetary structures. <https://bit.ly/417km01>

SpaceX launches ISRO's GSAT-20 satellite, aims to enhance India's communication infrastructure

In a significant development for India's space and communication sectors, SpaceX successfully launched the Indian Space Research Organisation's (ISRO) GSAT-20 communication satellite on Tuesday. The launch took place from Canaveral Space Force Station in Florida.

The GSAT-20 satellite, weighing 4,700 kg, is designed to strengthen India's communication infrastructure by supporting a wide range of services,



including broadband, in-flight internet, maritime communications, and cellular backhaul services. This advanced satellite is equipped with Ka-Ka band high throughput capacity and features 32 beams that will provide pan-India coverage, even extending to areas such as the Andaman and Nicobar Islands and Lak-

shadweep. The satellite is expected to operate for 14 years, improving communication across the country. <https://bit.ly/4i5IDuZ>